

The Agent Harness as Extended Phenotype

Multilevel Selection, Governed Transfer, and the Hidden Cost of Improving AI Agents

Ignacio Adrián Lerer

Independent Researcher

Attorney (UBA), Executive MBA (IAE, Universidad Austral)

ORCID: 0009-0007-6378-9749

Buenos Aires, Argentina

29 August 2026

Abstract

The effective capability of a language-model agent is not determined by model weights alone. Prompts, skills, memory, tools, verifiers, and control flow form a harness that regulates which latent capacities become operational behavior. DarwinX demonstrates that a frozen model can improve through population selection over harness variants. This article develops a bounded bridge between that engineering result, extended phenotype theory, and evolutionary game theory, then tests the bridge in a deterministic mechanism pilot. Three selection policies receive identical mutation streams: greedy single-lineage selection, capability-oriented population selection with recombination, and an EPT/EGT-inspired multilevel policy with non-compensable preservation gates, vertical specialization, ecosystem complementarity, and receiver-specific horizontal transfer. Across forty heterogeneous and forty homogeneous landscapes, multilevel selection did not improve aggregate or held-out coverage. In heterogeneous landscapes, its mean held-out coverage was 0.822, compared with 0.941 for both permissive policies. It nevertheless reduced accepted protected-capability regressions and compliance-failure runs to zero, at approximately twice the evaluation work of single-lineage selection. The result is not evidence that extended phenotype theory or evolutionary game theory has been empirically validated in AI systems. It is evidence that changing the unit, level, and admissibility conditions of selection changes what improvement means. A selector can obtain lower visible performance while preserving properties that an aggregate benchmark permits local adaptations to destroy. The paper therefore reframes harness evolution as a problem of constrained institutional selection rather than unconstrained score maximization.

Keywords: AI agents; agent harness; extended phenotype; evolutionary game theory; multilevel selection; horizontal transfer; governance; compliance; reproducibility.

1. Introduction

Comparisons among language models often treat an agent as if it were a direct expression of model weights. The identification is incomplete. Two agents using the same frozen model can behave differently when their instructions, tools, memories, retry policies, verification rules, or stopping conditions differ. Effective capability belongs to the composed system. The model supplies dispositions; the harness determines which dispositions can be expressed, under which conditions, and with what evidence of success.

DarwinX makes this observation operational. It evolves a population of harnesses while holding the base model fixed. Candidate edits may affect prompts, skills, memory, tools, or control flow. Variants survive according to measured performance, alternative lineages remain available for later recombination, and a preserve-and-extend contract limits regression. The authors report gains across in-domain, held-out, synthetic-to-real, and cross-benchmark settings. Their central contribution is not another prompt optimizer. It is a population-level selection process over persistent, inspectable scaffolds.

That achievement leaves a governance problem open. Preservation is never neutral. A verifier, benchmark, tolerance, or probe defines which properties count as fitness and which losses remain invisible. If all dimensions enter a scalar score, a large gain can compensate for the destruction of a protected capability. If some dimensions become hard admissibility conditions, the feasible frontier changes. The resulting selector may appear weaker under aggregate performance while becoming more reliable under a richer account of acceptable behavior.

This paper asks three questions. First, under what conditions may an agent harness be treated as an extended phenotype rather than merely as external software? Second, what additional structure is required before the language of evolutionary game theory becomes analytically useful? Third, what happens when selection operates simultaneously at individual, vertical, and ecosystem levels and makes protected behavior and receiver compatibility non-compensable?

The contribution is deliberately limited. The paper proposes an operational bridge and evaluates its mechanism in a deterministic synthetic pilot. It does not reproduce DarwinX, train a model, or claim biological identity between organisms and AI systems. Its principal finding is negative but informative: the multilevel policy failed to improve held-out coverage. It eliminated protected regressions and compliance failures, but did so at a substantial evaluation cost. Harness evolution therefore presents a capability-governance tradeoff, not a free improvement theorem.

2. From Frozen Models to Operational Phenotypes

2.1 The harness as a causal environment

An agent harness includes the persistent structures that mediate between model output and effective action: system and project instructions, tool descriptions, skill documents, retrieval and memory policies, validators, authorization gates, retry logic, and stopping rules. These structures do more than provide context. They shape the action set, alter the probability that a capability will be expressed, preserve procedural adaptations across tasks, and determine which outputs are accepted as completed work.

The resulting behavior can be represented schematically as:

$$O = F(M, H, T, E),$$

where O is observed performance, M is the base model, H is the harness, T is the task distribution, and E is the evaluation and execution environment. Holding M fixed does not hold O fixed because H changes the causal pathway from model disposition to operational consequence.

This formulation also prevents an overstatement. Externality alone does not make a component an extended phenotype. The analogy becomes useful only where the harness exhibits variation, persistence, differential retention, and transmissibility. A temporary prompt improvised for one exchange may be external context without belonging to an evolutionary lineage. A versioned skill that survives selection, is inherited by later agents, and changes their measurable effects has a stronger claim to extended-phenotype treatment.

2.2 What is inherited

In biological evolution, inheritance is constrained by reproduction. In agent systems, persistence can occur through repositories, configuration, memory stores, skill registries, benchmark suites, and deployment policies. These artifacts may outlive the model version that first generated or used them. They can migrate across model generations and organizational units. The relevant inheritance channel is institutional and computational rather than genetic.

This difference is not an objection to the analogy; it is a boundary condition. The selected object is not a biological organism. It is a socio-technical configuration that persists because an institution stores, evaluates, and redeploys it. Selection is therefore designed twice: first by the candidate-generation process and second by the evaluator that determines survival.

2.3 Reciprocal causation and constructed niches

The relationship is bidirectional. Models internalize regularities from training environments into latent parameters. Harnesses externalize adaptive structure into the task environment by constraining how models perceive, decide, act, and verify. Once deployed, those harnesses also modify the selective environment for later variants. A regression suite created after one failure changes which future mutations can survive. In this limited sense, harness engineering resembles niche construction: the selected system partly constructs the conditions of its own subsequent selection.

The analogy should remain disciplined. A benchmark does not become nature, and a model does not acquire biological interests. The useful claim is causal and institutional: persistent external structures alter behavior and the future retention of variants.

3. From Evolutionary Computation to Evolutionary Game Theory

DarwinX contains population, mutation, inheritance, recombination, and differential survival. It is therefore reasonably described as evolutionary computation. Evolutionary game theory requires more. Its distinctive questions concern strategic interaction, frequency-dependent payoffs, invasion, stability, and the distribution of gains and externalities among interacting strategies.

An agent-harness population begins to invite an EGT analysis when variants compete for deployment, evaluation budget, tool access, or institutional trust; when the success of one specialist depends on the presence of complementary specialists; when one adaptation shifts risks to another vertical; or when a verifier-exploiting mutation can invade a population of more reliable harnesses.

Let a candidate harness h have a capability vector $C(h)$, a protected-property vector $P(h)$, a compliance state $L(h)$, a vertical contribution $V(h)$, an ecosystem contribution $X(h)$, and evaluation cost $K(h)$. A scalar selector may maximize:

$$F_{\text{scalar}}(h) = w_C C(h) + w_V V(h) + w_X X(h) - w_K K(h).$$

Under that formulation, sufficient capability gain can compensate for a decline in P or L . The policy tested here instead uses constrained selection:

maximize $F_{\text{capability}}(h)$

subject to $P_j(h) \geq \tau_j$ for every protected property j ,

$L(h) \geq \lambda$,

and $R(h, v) = 1$ before transferring an adaptation to receiving vertical v .

The difference is normative as well as mathematical. Weights express tradeoffs. Gates express prohibitions on compensation. Neither is supplied by evolution itself. Both are institutional choices.

4. Multilevel Selection and Governed Horizontal Transfer

The proposed policy operates at three levels.

At the individual level, a harness mutation must add measured training capability without crossing protected-capability or compliance thresholds. This prevents an apparently beneficial child from surviving by sacrificing a property defined as non-compensable.

At the vertical level, specialists compete within heterogeneous task ecologies. A legal research adaptation need not defeat a software-engineering adaptation on one global scalar score. Each can retain locally useful structure, provided it satisfies the common preservation contract.

At the ecosystem level, selection rewards complementary coverage and diversity. A population is valuable not merely because it contains the highest-scoring individual, but because its members jointly preserve capabilities across task families.

Horizontal transfer connects the levels. A mutation discovered in one vertical may be offered to another, but its effects count only when the receiver passes a compatibility test. This implements a principle that is familiar in legal transplantation and organizational learning: a practice successful in one institutional environment cannot be presumed beneficial in another merely because its text can be copied.

The receiver-specific test also distinguishes transmission from adoption. A shared repository makes an adaptation available. It does not establish that every recipient should incorporate it. The possible outcomes are adopt, adapt, reject, or defer pending evidence.

5. Method

5.1 Design and preregistered hypotheses

The mechanism pilot was specified before execution. It compared three policies under identical mutation streams and local deterministic computation:

1. Single lineage: greedy retention when scalar training fitness increases.
2. Population: an archive with bounded visible regression and recombination of complementary variants.
3. EPT/EGT multilevel: individual preservation, vertical contribution, ecosystem complementarity, and governed transfer.

Four hypotheses were frozen. H1 predicted greater held-out ecosystem coverage for multilevel selection than for a single lineage. H2 predicted fewer protected-capability regressions than capability-oriented population selection. H3 predicted that any multilevel advantage would be largest in heterogeneous landscapes. H4 predicted higher evaluation work and therefore required reporting cost rather than performance alone.

5.2 Synthetic landscapes

Forty seeds generated heterogeneous landscapes and forty generated homogeneous landscapes. Each landscape produced an ordered stream of seventy-two mutations. All three policies received the same stream for a given seed. The state contained sixteen training capabilities, eight held-out capabilities, four protected capabilities, a compliance value, accepted-mutation history, vertical origins, and evaluation work.

Each mutation had a vertical of origin, gains, possible local losses, possible protected-capability losses, a compliance delta, an evaluation cost, a transmissibility flag, and a receiver-compatibility flag. Heterogeneous landscapes distributed origins among harness, research, security, and legal verticals. Homogeneous landscapes confined origin to one vertical.

The values and probabilities were generated by a seeded pseudorandom process. The simulation does not estimate these values from observed agents. They define a controlled landscape for comparing selectors, not an empirical model of real-world mutation frequency.

5.3 Separation of selection and evaluation

Held-out capability was never used to promote a mutation. The single-lineage policy accepted a child when training fitness increased. The population policy retained bounded-regression children in an archive and evaluated recombinations after mutation search. The multilevel policy evaluated candidates against a vertical specialist and the current ecosystem leader, required training improvement and non-decreasing vertical coverage, and rejected any child below the protected or compliance gates. Transfer effects materialized only when the mutation was transferable and receiver-compatible.

5.4 Outcome measures

The recorded outcomes were training coverage, held-out coverage, accepted protected-capability regressions, runs ending below the compliance threshold, lineage diversity, accepted mutations, and evaluation work. The work measure is an abstract count based on candidate costs and policy-specific comparisons. It is not a monetary estimate.

5.5 Reproducibility boundary

The pilot used no model inference, private data, or production self-modification. The source script, frozen hypotheses, raw per-seed observations, aggregates, tests, manifest, and provenance receipt form the reproducibility package. Reproduction establishes that the reported numbers follow from the disclosed simulator. It does not validate the simulator as a model of deployed AI systems.

6. Results

6.1 Heterogeneous landscapes

Single-lineage and population selection each reached mean training coverage of 0.989 and held-out coverage of 0.941. Multilevel selection reached 0.959 training coverage and 0.822 held-out coverage. H1 was not supported.

The difference appeared in protected properties. Single-lineage selection accepted a mean of 3.225 protected regressions per run and population selection accepted 3.100. Multilevel selection accepted none. Runs ending below the compliance threshold averaged 0.975 for each permissive policy and zero for multilevel selection. H2 was supported within the simulator.

Evaluation work averaged 179.85 for single lineage, 299.85 for population selection, and 363.70 for multilevel selection. The preservation result was therefore not free.

6.2 Homogeneous landscapes

All three policies reached training coverage of 0.250. Held-out coverage was 0.069 for the two permissive policies and 0.044 for multilevel selection. Protected regressions averaged 1.275 for single lineage, 0.075 for population selection, and zero for multilevel selection. Compliance-failure runs averaged 0.425 for the permissive policies and zero for multilevel selection. Evaluation work was 178.68, 224.78, and 361.35 respectively.

H3 received only partial support as a boundary observation. Heterogeneity raised absolute coverage for all selectors, but it did not create a performance advantage for the multilevel policy. H4 was supported: multilevel selection required approximately twice the evaluation work of single-lineage selection.

6.3 Hypothesis disposition

H1, superior held-out coverage, was rejected. H2, fewer protected regressions, was supported in the simulated mechanism. H3, a larger multilevel advantage under heterogeneity, was not supported as a performance claim. H4, higher evaluation cost, was supported.

7. Discussion

7.1 The negative result is the contribution

The simplest evolutionary narrative predicts that more structured selection will produce a stronger agent. The pilot contradicts that narrative. The multilevel policy obtained less measured capability because it rejected some mutations that the permissive selectors retained. Those mutations were beneficial under scalar training fitness but inadmissible under protected-property, compliance, or receiver-compatibility constraints.

The result does not show that the multilevel policy is better in the abstract. It shows that selectors answer different questions. A capability selector asks which candidate increases measured performance. A governed selector asks which candidate increases performance inside a constrained set of acceptable states. The second may rank lower on the first selector's metric without malfunctioning.

7.2 Fitness is an institutional constitution

In deployed systems, benchmarks, regression suites, approval rules, and compliance probes act like a constitution for artificial selection. They determine what may be exchanged for what. Treating safety or legality as a weighted penalty allows compensation. Treating them as gates forbids it. This distinction should be explicit whenever an organization claims that an agent has improved.

The problem is particularly acute in legal and regulated settings. A drafting agent might become faster while losing source traceability. A research agent might retrieve more authorities while becoming less reliable about jurisdiction. A workflow agent might complete more tasks by exceeding delegated authority. Aggregate success can conceal these degradations unless the relevant properties are measured and non-compensable.

7.3 Transfer is not copying

The pilot's receiver-compatibility condition operationalizes a broader principle. Harness artifacts are portable, but their fitness is environment-dependent. A concise research prompt can improve one model and degrade another. A verification skill designed for public web research may be unsafe for a confidential matter. A scheduler that benefits one workflow may create duplicate or unauthorized actions in another.

Governed horizontal transfer therefore requires evidence from the receiver. The adaptation should carry provenance, source conditions, expected benefit, protected properties, test results, and a rollback path. Shared memory should transmit candidate knowledge, not automatic authority.

7.4 Relation to DarwinX

DarwinX already recognizes bounded regression, preservation probes, heterogeneous specialists, and recombination. Its later discussion expressly observes that compliance probes could become selection constraints, while describing the general version as untested. The present pilot isolates that move and makes protected behavior and receiver compatibility non-compensable. It then measures the consequence: preservation improves, visible coverage declines, and evaluation work rises.

The studies are not competing replications. DarwinX evaluates real agent harnesses across demanding benchmarks and reports capability gains. This paper evaluates a synthetic selector and reports a governance tradeoff. Its purpose is to expose a design decision that high benchmark performance cannot settle by itself.

7.5 Stability and invasion

An EGT extension would test whether reliable harnesses remain stable against opportunistic variants. A mutation that exploits the verifier may obtain high apparent fitness while reducing real task validity. A system with only aggregate selection may permit such a variant to invade. A preservation-and-authority architecture should detect and exclude it, but overly strict gates can freeze innovation.

Future experiments should therefore vary proposer diversity, verifier noise, frequency-dependent interaction, deceptive mutations, transfer compatibility, and gate strictness. The central outcome should be a Pareto frontier among capability, protected behavior, cost, and adaptability, not one universal score.

8. Legal and Governance Implications

The harness is a natural unit of legal and organizational inquiry because it contains many controls that determine effective conduct: permissions, tool access, instructions, logging, verification, escalation, and stopping. Liability and compliance analysis that focuses only on the foundation model can miss the institutional structures that made a harmful output actionable or allowed it to persist.

This does not mean that the harness bears legal responsibility. Responsibility remains attached to persons and organizations under applicable law. The point is evidentiary and architectural. A versioned harness can reveal which rules applied, which tools were available, what evidence justified a change, and whether an operator knowingly accepted a regression. It may therefore become part of the record through which reasonable design, governance, delegation, and oversight are assessed.

For legal practice, four operational implications follow. First, agent evaluation should preserve jurisdiction, authority, confidentiality, and citation integrity as explicit constraints. Second, improvements discovered in one practice area should not automatically propagate to another. Third, every promoted change should leave a readable diff and evidence receipt. Fourth, autonomous external action should remain behind authorization controls that cannot be compensated by higher task performance.

These are governance propositions, not statements of a particular jurisdiction's positive law. Their legal significance depends on the applicable regulatory, professional, contractual, and evidentiary framework.

9. Limitations

The landscapes are synthetic. Mutation effects, transfer relations, costs, and compliance impacts were generated rather than observed. No language model proposed or executed a mutation. The verifiers were deterministic and therefore not exposed to reward hacking, correlated model error, or evaluator drift. The policies did not receive equal evaluation work, although they received identical candidate streams; the extra work is an outcome of the policy design rather than a controlled compute-equivalent comparison.

The zero-regression result is partly structural because the multilevel selector is programmed to reject states below the protected thresholds. The empirical question is not whether a correctly implemented gate rejects a disclosed violation. It is whether real probes detect the relevant regression, whether protected properties have been specified adequately, and whether the resulting benefit justifies their cost and false-rejection rate.

The bridge to extended phenotype theory remains analogical and mechanistic, not ontological. The pilot does not validate EPT or EGT in artificial systems. It uses those theories to generate distinctions about inheritance, levels, externalities, transmission, and stability that can be operationalized and falsified.

10. Next Experiment

The next phase should be preregistered and use public or synthetic agent tasks. It should freeze the base model, prompts, task suite, tool environment, and budget ceiling. Candidate mutations should be proposed under the same process for every arm. Critical constraints should be checked deterministically where possible, with blinded human or independent adjudication for ambiguous cases.

At minimum, the experiment should compare scalar, bounded-regression, and non-compensable selectors; include opportunistic mutations; measure task success, protected regressions, transfer success, false rejection, cost per acceptable output, and recovery after a model swap; and publish null results. The live-agent phase should remain blocked until those conditions are fixed.

11. Conclusion

A frozen model need not be a fixed agent because durable capability can reside in the harness around it. That observation supports a bounded extended-phenotype interpretation when harness adaptations vary, persist, transmit, and undergo differential retention. Evolutionary game theory adds questions of interaction, externality, invasion, and stability that ordinary evolutionary optimization does not answer.

The mechanism pilot did not show that EPT/EGT-inspired selection improves aggregate performance. It showed that a non-compensable selection constitution can eliminate protected regressions and compliance failures that score-oriented selectors accept, while sacrificing coverage and consuming more evaluation work. The correct question is therefore not whether evolution makes an agent better. It is which properties the selection system permits an apparent improvement to destroy.

Epistemic Status

This paper treats the EPT/EGT bridge as a falsifiable auxiliary hypothesis within a progressive research programme. The negative performance result is retained rather than repaired through retrospective redefinition. Source evidence, theoretical extension, and simulated mechanism evidence are distinguished throughout.

AI Use Disclosure

Artificial intelligence tools assisted with source extraction, protocol design, programming, analysis, translation, and drafting. The author defined the research question, approved the conceptual framework, reviewed the evidence, and remains responsible for the hypotheses, interpretations, claims, and publication decision. The experiment was executed locally through deterministic code, and its source, results, and tests were retained for review.

Data and Code Availability

The reproducibility package accompanying the Zenodo record contains the deterministic simulator, frozen hypotheses, method specification, raw and aggregate results, tests, manifest, and provenance receipt. No private or personal data were used.

Conflicts of Interest and Funding

The author declares no conflict of interest relevant to this study. No external funding supported the reported pilot.

References

- Campbell, D. T. (1974). Evolutionary epistemology. In P. A. Schilpp (Ed.), *The Philosophy of Karl Popper* (pp. 413-463). Open Court.
- Dawkins, R. (1982). *The Extended Phenotype: The Long Reach of the Gene*. Oxford University Press.
- Hull, D. L. (1988). *Science as a Process*. University of Chicago Press.
- Lakatos, I. (1978). *The Methodology of Scientific Research Programmes*. Cambridge University Press.
- Maynard Smith, J. (1982). *Evolution and the Theory of Games*. Cambridge University Press.
- Odling-Smee, F. J., Laland, K. N., & Feldman, M. W. (2003). *Niche Construction: The Neglected Process in Evolution*. Princeton University Press.
- Okasha, S. (2006). *Evolution and the Levels of Selection*. Oxford University Press.
- Popper, K. R. (1959). *The Logic of Scientific Discovery*. Hutchinson.
- Yang, Y., Gong, Z., Huang, W., Yang, Q., Zhou, Z., Huang, Z., Li, Y., Gao, X., Dai, Q., Liu, B., Qiu, K., Yang, Y., Chen, D., Yang, X., & Luo, C. (2026). SkillOpt: Executive strategy for self-evolving agent skills. arXiv:2605.23904.
- Zhang, J., Hu, S., Lu, C., Lange, R., & Clune, J. (2025). Darwin Gödel Machine: Open-ended evolution of self-improving agents. arXiv:2505.22954.
- Zhang, Y., Dai, Y., Tan, J., Yang, L., Mullur, R., Hoang, T., Hu, Z., Zhu, J., Mui, P., Savarese, S., Xu, R., & Chen, Z. (2026). DarwinX: Evolving Agent Harnesses Through Natural Selection. arXiv:2608.07545v1. <https://doi.org/10.48550/arXiv.2608.07545>